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Buck Converter

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Date: June, 2026

1. Executive Summary

This report documents the complete design of a non-isolated step-down (buck) DC-DC converter that produces a regulated 5 V, 3 A output (15 W) from a nominal 12 V input. The converter is built around the Texas Instruments TPS54360, a peak current-mode controller with an integrated high-side MOSFET, in an asynchronous configuration using an external Schottky catch diode.

The design was carried out from first principles: every component value is derived from the governing datasheet equations and a worst-case operating-point analysis, rather than copied from a reference design. The work covers the full power stage (inductor, input and output capacitors, catch diode), the regulation and protection networks (feedback divider, frequency-set resistor, undervoltage lockout), and a Type II compensation network designed to a specific loop target.

Key results:

- **Output regulation** of 4.984 V (0.3% error) against the 5 V ±2% target.
- **Output ripple** below the 50 mV specification, with the output capacitor bank sized by the load-transient requirement.
- **Control loop** crossing over near 50 kHz with approximately 62° of phase margin, confirmed by both a hand calculation and a numerical Bode analysis, predicting a stable, well-damped response.
- **Predicted efficiency** of 89.7% at full load, with the loss budget identifying the Schottky diode as the dominant loss and synchronous conversion as the clearest path to improvement.

The design was captured as a schematic and a four-layer PCB layout in KiCad, both passing their respective rule checks (ERC and DRC). The project is a complete paper design verified by analysis; building and bench-testing a prototype is identified as the primary next step.

2. Specifications

The converter was designed to the following target specifications, representative of a point-of-load regulator supplying a 5 V rail from a 12 V bus.

Parameter	Value
Input Voltage (Vin)	10.8V - 13.2V (12V nominal)
Output Voltage (Vout)	5V ±2%
Output Current (Iout)	0 - 3A
Output Ripple	<50mVpp
Transient deviation	±200mV (4% of 5V), load step 0.5A → 3A
Switching frequency (fsw)	500kHz
Topology	Asynchronous buck (HS FET + external Schottky)
Target Efficiency	>90% at full load

The ±10% input range reflects a typical regulated 12 V bus with tolerance. The 500 kHz switching frequency was selected as a balance between component size and switching loss, discussed in Section 3. The transient specification of ±200 mV drove the output capacitor sizing, as the load-step requirement proved more demanding than the steady-state ripple requirement.

3. Topology Selections

This section explains the three main architectural choices: the converter topology, the rectification scheme, and the control method. Each choice involves a careful trade-off rather than being a default.

3.1 Why a Buck Converter

The application needs to step a 12 V input down to a 5 V output. A buck converter is the standard choice for this step-down conversion because it switches instead of dissipating power. A linear regulator could convert 12 V to 5 V, but it would drop the 7 V difference across a pass transistor, wasting energy as heat. At 3 A, that results in a loss of 21 W for 15 W of output, leading to an efficiency below 42%. In contrast, a switching buck converter transfers energy through an inductor with only minor conduction and switching losses, allowing for an efficiency above 90%. For a 15 W rail, this difference in efficiency is crucial.

3.2 Why Asynchronous (Schottky Diode) Rectification

A buck converter needs a path for the inductor current when the high-side switch is off. This path can be provided by a diode (asynchronous) or by a second, actively-driven low-side MOSFET (synchronous).

The TPS54360 integrates only the high-side MOSFET, so the design uses an external Schottky diode for the low-side path, making it asynchronous. This choice is suitable for the power level. At 3 A, the diode conduction loss is significant but manageable, and the asynchronous approach is simpler, requiring fewer components and eliminating concerns over low-side gate-drive timing or shoot-through. The trade-off is efficiency: as shown in the loss budget (Section 8), the Schottky diode is the largest source of loss. A synchronous design could improve efficiency and is identified as a key improvement for future work. For an initial power-electronics design at this level, the asynchronous topology offers the right balance of performance and simplicity.

A Schottky diode is chosen over a standard silicon rectifier because of its lower forward voltage, which reduces conduction loss, and its negligible reverse-recovery charge, which minimizes switching loss and electromagnetic interference at the 500 kHz switching frequency.

3.3 Why Peak Current-Mode Control

The TPS54360 uses peak current-mode control, where the inductor current is sensed and regulated cycle by cycle. This method is preferred over voltage-mode control for three reasons. First, it offers built-in cycle-by-cycle overcurrent protection since the controller directly monitors the peak inductor current. Second, it simplifies feedback compensation. Current-mode control removes the inductor from the small-signal model, making the power stage a single dominant pole instead of a two-pole LC response. This change allows for a simpler Type II compensator (designed in Section 5) rather than a Type III. Third, it provides good rejection of line transients, as input voltage changes are corrected within the inner current loop before they significantly affect the output.

3.4 Why 500 kHz

The switching frequency creates a trade-off between passive component size and switching losses. A higher frequency allows for a smaller inductor and smaller capacitors (with less energy stored per cycle), which reduces board area and cost but increases switching losses in the MOSFET and the gate-drive power, both of which rise with frequency. A lower frequency improves efficiency but leads to larger magnetics and capacitors. A switching frequency of 500 kHz is a practical compromise for this power level. It keeps the inductor compact at 8.2 µH while limiting switching losses to about 10% of the total loss budget. This frequency is also well within the TPS54360's operating range and maintains a significant margin between the switching frequency and the selected 50 kHz loop crossover.

4. Component Selection

This section documents the selection of every passive and discrete component in the converter. Each value is derived from first principles using the TPS54360 datasheet (SLVSBB4G) equations and the operating point established in Section 1, then mapped to the nearest available standard part. The design philosophy throughout is to size each component for worst-case conditions with documented margin, rather than for nominal operation.

4.1 Operating Point

All downstream component sizing depends on the converter's duty cycle and current waveforms, so these are established first.

The ideal duty cycle neglects all device drops:

$D_{ideal} = V_{out} / V_{in} = 5 / 12 = 0.417$ (at nominal Vin)

Because the TPS54360 is asynchronous (a high-side MOSFET plus an external Schottky diode), the real duty cycle must account for the high-side on-resistance and the Schottky forward voltage:

$$
\begin{aligned}
D &= (V_{out} + V_F + I_{out} \cdot R_{dc}) / (V_{in} - I_{out} \cdot R_{DS(on)} + V_F) \\
&= (5 + 0.5 + 0) / (12 - 3 \cdot 0.092 + 0.5) \\
&= 5.5 / 12.224 \\
&= 0.450
\end{aligned}
$$

The real duty cycle is therefore 0.450 at nominal input, about 8% higher than ideal because the Schottky drop is larger than a synchronous FET drop would be. Across the full input range the real duty cycle spans 0.499 (at Vin = 10.8 V) to 0.410 (at Vin = 13.2 V). The value D = 0.450 is carried forward into all loss and ripple calculations.

The switching period and on/off times follow directly:

$$
\begin{aligned}
T_{sw} &= 1 / f_{sw} = 1 / 500\ \text{kHz} = 2.0\ \mu\text{s} \\
t_{on} &= D \cdot T_{sw} = 0.450 \cdot 2.0\ \mu\text{s} = 900\ \text{ns} \\
t_{off} &= (1 - D) \cdot T_{sw} = 1100\ \text{ns}
\end{aligned}
$$

A minimum-on-time check confirms the converter can regulate at the highest input voltage, where on-time is shortest. At Vin = 13.2 V (D = 0.410):

$t_{on,min} = 0.410 \cdot 2.0\ \mu\text{s} = 820$ ns

Compared against the chip's hardware limit of 135 ns, this gives a margin of 820 / 135 = 6.1×. The converter operates comfortably within the controllable on-time range across the entire input voltage span.

4.2 Inductor

The inductor is the energy-storage element that smooths the chopped switch-node voltage into a continuous output current. Its value sets the peak-to-peak ripple current, which in turn affects output ripple, peak current stress, and losses.

A ripple target of 30% of maximum output current ($K_{IND} = 0.3$) is selected, following TI's recommendation for designs using ceramic output capacitors. This balances inductor size against output-capacitor requirements. The minimum inductance is computed at the worst case for ripple, which is maximum input voltage (longest off-time):

$$
\begin{aligned}
L_{min} &= V_{out} \cdot (V_{in,max} - V_{out}) / (I_{out} \cdot K_{IND} \cdot V_{in,max} \cdot f_{sw}) \\
&= 5 \cdot (13.2 - 5) / (3 \cdot 0.3 \cdot 13.2 \cdot 500{,}000) \\
&= 41 / 5{,}940{,}000 \\
&= 6.90\ \mu H
\end{aligned}
$$

The nearest standard value above this minimum is 8.2 µH, which also matches the value used in the TI datasheet design example — a useful independent check on the calculation.

With L = 8.2 µH, the actual ripple current is recomputed:

$$
\begin{aligned}
\Delta i_L &= V_{out} \cdot (V_{in,max} - V_{out}) / (V_{in,max} \cdot L \cdot f_{sw}) \\
&= 41 / 54.12 \\
&= 0.757\ A
\end{aligned}
$$

This is 25.2% of the 3 A full load, comfortably below the 30% target. The peak and RMS inductor currents follow:

$$I_{L,peak} = I_{out} + \Delta i_L/2 = 3 + 0.379 = 3.38\ A$$
$$I_{L,RMS} = \sqrt{I_{out}^2 + \Delta i_L^2/12} = \sqrt{9 + 0.048} = 3.01\ A$$

The peak current must remain below the chip's current limit. Against the guaranteed minimum current limit of 4.5 A, the margin is 4.5 / 3.38 = 1.33×, which is acceptable.

The selected part is the Coilcraft XAL7030-822MEC: 8.2 µH, saturation current 8.9 A, RMS rating 6.4 A, and a maximum DC resistance of 19.1 mΩ. The saturation margin (8.9 A versus the 4.5 A chip current limit) is roughly 2×, ensuring the inductor cannot saturate even under a fault. The DCR contributes a conduction loss of $I_{L,RMS}^2 \cdot DCR = 3.01^2 \cdot 0.0191 = 0.173$ W at full load.

4.3 Output Capacitor

The output capacitor performs three jobs simultaneously: filtering the inductor ripple current into a smooth output voltage, supplying transient current during a sudden load increase, and absorbing energy during a sudden load decrease. It must satisfy all three criteria, and the most demanding one governs the value.

Loading transient. When the load steps up, the control loop takes a few switching cycles to respond; the capacitor must supply the deficit current without the output sagging beyond the allowed deviation:

$$
\begin{aligned}
C_{OUT,1} &\geq 2 \cdot \Delta I_{out} / (f_{sw} \cdot \Delta V_{out}) \\
&= 2 \cdot 2.5 / (500{,}000 \cdot 0.2) \\
&= 50\ \mu F
\end{aligned}
$$

Unloading transient. When the load steps down, the inductor's stored energy charges the capacitor, causing overshoot. By energy conservation:

$$
\begin{aligned}
C_{OUT,2} &\geq L \cdot (I_{OH}^2 - I_{OL}^2) / (V_f^2 - V_i^2) \\
&= 8.2\mu H \cdot (3^2 - 0.5^2) / (5.2^2 - 5^2) \\
&= 8.2\mu H \cdot (8.75 / 2.04) \\
&= 35.2\ \mu F
\end{aligned}
$$

Steady-state ripple. Even at constant load, the inductor ripple current produces a small output ripple:

$$
\begin{aligned}
C_{OUT,3} &\geq \Delta i_L / (8 \cdot f_{sw} \cdot V_{ripple}) \\
&= 0.757 / (8 \cdot 500{,}000 \cdot 0.050) \\
&= 3.78\ \mu F
\end{aligned}
$$

The loading transient dominates at 50 µF of required effective capacitance. The ESR limit and RMS current are also checked:

$$R_{ESR} \leq V_{ripple} / \Delta i_L = 0.050 / 0.757 = 66\ m\Omega$$
$$I_{COUT,RMS} = \Delta i_L / \sqrt{12} = 0.757 / 3.464 = 219\ mA$$

Ceramic capacitors lose capacitance under DC bias, typically around 30% at this voltage. To achieve 50 µF effective, the nameplate value must be at least 50 / 0.7 = 71 µF. Two 47 µF capacitors in parallel provide 94 µF nameplate, or about 66 µF effective after derating — a 32% margin over requirement. The selected part is the Murata GRM32ER61C476ME15L (47 µF, 16 V, X5R, 1210). Its ESR of a few milliohms is far below the 66 mΩ limit, and its RMS rating exceeds 1 A per cap against the 110 mA each actually carries.

4.4 Input Capacitor

The input capacitor supplies the pulsed current the converter draws from the source each switching cycle. Because the high-side switch only conducts during the on-time, the input current is a pulse train rather than a smooth DC draw. This makes the dominant design constraint the RMS current rating, not capacitance.

The RMS input ripple current is worst at the input voltage that produces a duty cycle of 0.5, which occurs at Vin = 10.8 V:

$$
\begin{aligned}
I_{CIN,RMS} &= I_{out}\cdot\sqrt{(V_{out}/V_{in})\cdot((V_{in} - V_{out})/V_{in})} \\
&= 3\cdot\sqrt{0.463 \cdot 0.537} \\
&= 3\cdot\sqrt{0.2486} \\
&= 1.50\ \text{A}
\end{aligned}
$$

The required capacitance, for a target input ripple of 250 mV:

$$
\begin{aligned}
C_{IN} &\geq I_{out}\cdot 0.25 / (f_{sw} \cdot \Delta V_{in}) \\
&= 3\cdot 0.25 / (500{,}000 \cdot 0.25) \\
&= 6\ \mu\text{F}
\end{aligned}
$$

After DC-bias derating, the nameplate requirement is 6 / 0.75 = 8 µF. Two 4.7 µF capacitors in parallel give 9.4 µF nameplate, about 7 µF effective. Each cap carries 750 mA RMS against a rating of roughly 3 A, a 4× margin. The selected part is the Murata GRM32ER71E475KA88L (4.7 µF, 25 V, X7R, 1210). The 25 V rating provides 1.9× margin over the maximum input voltage, and the X7R dielectric is chosen over Y5V for its far smaller capacitance loss with temperature and bias.

4.5 Bootstrap Capacitor

The bootstrap capacitor generates the gate-drive voltage for the high-side N-channel MOSFET, which requires a gate voltage several volts above the switch node. While the switch node is low, the chip charges this capacitor; when the switch node rises to the input voltage, the capacitor rides up with it, maintaining the elevated gate drive.

The datasheet specifies a fixed value of 100 nF with an X5R or X7R dielectric rated at 10 V or higher. No calculation is required — this value is chosen by the chip designers based on the internal FET gate charge. The selected part is the Murata GCM188R71E104KA37D (100 nF, 25 V, X7R, 0603), the 25 V rating giving ample margin.

4.6 Feedback Divider

The converter regulates the output by comparing a divided fraction of the output voltage against an internal 0.8 V reference at the FB pin. The lower resistor is chosen first, then the upper resistor is derived.

R_{LS} = 10 kΩ is selected as a standard value that draws 80 µA through the divider — a balance between quiescent loss (favoring larger resistors) and noise immunity at the high-impedance FB pin (favoring smaller resistors). The upper resistor follows from the datasheet divider equation:

$$
\begin{aligned}
R_{HS} &= R_{LS}\cdot(V_{out} - 0.8)/0.8 \\
&= 10\text{k}\Omega\cdot(4.2/0.8) \\
&= 52.5\ \text{k}\Omega
\end{aligned}
$$

The nearest E96 1% standard value is 52.3 kΩ. Recomputing the actual output voltage with this value:

$V_{out} = 0.8 \cdot (1 + 52.3/10) = 4.984$ V

This is a 0.3% error, well inside the ±2% specification. Both resistors are 1% metal-film parts, since output-voltage accuracy depends directly on the divider ratio.

4.7 Frequency-Set Resistor

A single resistor from the RT/CLK pin to ground sets the switching frequency, per the datasheet equation:

$$R_T = 101{,}756 / f_{sw}^{1.008}$$
$$= 101{,}756 / 500^{1.008}$$
$$= 193.7\ k\Omega$$

The nearest E96 standard value is 191 kΩ. The resulting actual frequency is verified with the inverse equation:

$$f_{sw} = 92{,}417 / 191^{0.991} = 502\ kHz$$

This is 0.4% above the 500 kHz target, an acceptable deviation. A 1% resistor is used and placed close to the RT/CLK pin, which the datasheet flags as noise-sensitive.

4.8 UVLO Divider

The undervoltage-lockout divider on the EN pin keeps the converter disabled until the input voltage is high enough for proper operation, and sets distinct turn-on and turn-off thresholds for hysteresis. Targets of 9.0 V start and 8.0 V stop are chosen, both safely below the 10.8 V minimum operating input, with 1 V of hysteresis to prevent chattering near the threshold.

The upper resistor is set by the hysteresis current:

$$R_{UVLO1} = (V_{START} - V_{STOP}) / I_{HYS}$$
$$= 1.0 / 3.4\mu A$$
$$= 294\ k\Omega$$

The lower resistor follows from the enable threshold and pull-up current:

$$R_{UVLO2} = V_{ENA} / [(V_{START} - V_{ENA})/R_{UVLO1} + I_1]$$
$$= 1.2 / [(7.8/294{,}000) + 1.2\mu A]$$
$$= 1.2 / 27.7\mu A$$
$$= 43.3\ k\Omega$$

The nearest standard values are 294 kΩ and 43.2 kΩ, both 1% parts.

4.9 Schottky Catch Diode

When the high-side FET turns off, the inductor current must continue to flow; the Schottky diode provides this path from ground to the switch node. A Schottky is chosen over a silicon rectifier for its low forward voltage (reducing conduction loss) and negligible reverse-recovery (reducing switching loss and EMI).

The requirements are a reverse rating above the maximum input voltage with margin (60 V chosen against 13.2 V max), a forward current rating above the output current (5 A chosen against 3 A), and the lowest practical forward voltage. The selected part is the B560C-13-F: 60 V reverse, 5 A forward, with a forward voltage of 0.55 V at 3 A. Its conduction loss at full load, accounting for the fraction of the cycle it conducts, is:

$$P_D = I_{out} \cdot V_F \cdot (1 - D) = 3 \cdot 0.55 \cdot 0.55 = 0.91\ W$$

This is the single largest loss in the converter, and is the principal reason a synchronous topology would improve efficiency (discussed in Section 11).

4.10 Summary

The table below collects every selected component. The compensation network (R_COMP, C_COMP, C_HF) is derived separately in Section 5, as its values depend on the control-loop analysis rather than the power-stage operating point.

Ref	Component	Value	Part Number
U1	Controller	N/A	TPS54360DDA
L1	Inductor	8.2 µH	Coilcraft XAL7030-822MEC
C_IN1, C_IN2	Input caps	4.7 µF 25V X7R	Murata GRM32ER71E475KA88L
C_OUT1,C_OUT2	Output caps	47 µF 16V X5R	Murata GRM32ER61C476ME15L
C_BOOT	Bootstrap	100 nF 25V X7R	Murata GCM188R71E104KA37D
D1	Schottky	60V 5A	B560C-13-F
R_HS	FB top	52.3 kΩ 1%	Vishay CRCW060352K3FKEA
R_LS	FB bottom	10.0 kΩ 1%	Vishay CRCW060310K0FKEA
R_T	Freq set	191 kΩ 1%	Vishay CRCW0603191KFKEA
R_UVLO1	UVLO top	294 kΩ 1%	CRCW0603294KFKEA
R_UVLO2	UVLO bottom	43.2 kΩ 1%	CRCW060343K2FKEA

5. Control Loop Design

This section designs the feedback compensation that makes the converter stable and responsive. The approach follows standard practice for a current-mode buck: first model the power stage as a transfer function (the "plant"), then design a compensator that shapes the overall loop to cross unity gain at a chosen frequency with adequate phase margin. The TPS54360 uses peak current-mode control, which simplifies this task because the inductor is effectively removed from the small-signal model, leaving a single-pole plant.

5.1 Why Current-Mode Control Simplifies the Problem

In voltage-mode control, the inductor and output capacitor form a two-pole LC filter, producing a 180-degree phase drop that demands a more complex Type III compensator to stabilize. Peak current-mode control instead regulates inductor current cycle by cycle, so the inductor behaves as a controlled current source rather than part of an LC resonance. The small-signal plant collapses to a single dominant pole set by the output capacitor and load, plus a high-frequency zero from the capacitor's ESR. This single-pole plant can be compensated with a Type II network (one integrator pole, one zero, one high-frequency pole), which is simpler to design and more robust than Type III.

5.2 Plant Transfer Function

The power stage transfer function, from the control voltage at the COMP pin to the output voltage, takes the form given in the datasheet:

$$V_{out}/V_c(s) = A_{dc} \cdot (1 + s/2\pi \cdot f_Z) / (1 + s/2\pi \cdot f_P)$$

Three quantities define it: the DC gain A_{dc}, the dominant pole f_P, and the ESR zero f_Z.

DC gain. The load resistance at full load is:

$$R_L = V_{out} / I_{out} = 5 / 3 = 1.667\ \Omega$$

The DC gain is the product of the power-stage modulator transconductance ($g_{mps} = 12$ A/V for this device) and the load resistance:

$$A_{dc} = g_{mps} \cdot R_L = 12 \cdot 1.667 = 20.0\ \text{V/V} = 26\ \text{dB}$$

Dominant pole. The output capacitor working into the load resistance sets the single dominant pole. Using the effective output capacitance of 66 µF:

$$
\begin{aligned}
f_P &= 1 / (2\pi \cdot R_L \cdot C_{out}) \\
&= 1 / (2\pi \cdot 1.667 \cdot 66\mu F) \\
&= 1447 \text{ Hz}
\end{aligned}
$$

ESR zero. The output capacitor's equivalent series resistance forms a zero with the capacitance. With a ceramic ESR of approximately 1.5 mΩ:

$$
\begin{aligned}
f_Z &= 1 / (2\pi \cdot R_{ESR} \cdot C_{out}) \\
&= 1 / (2\pi \cdot 1.5m\Omega \cdot 66\mu F) \\
&= 1.61 \text{ MHz}
\end{aligned}
$$

Because ceramic capacitors have very low ESR, this zero sits at 1.61 MHz, far above the crossover frequency. It therefore has negligible effect on loop shape in the frequency range of interest — an advantage of ceramic output capacitors over electrolytics, whose lower-frequency ESR zeros complicate compensation.

Plant behavior at the target crossover. A crossover frequency of 50 kHz is chosen (discussed in 5.3). Evaluating the plant there, well above the pole and below the zero, the gain rolls off at −20 dB/decade:

$$
\begin{aligned}
\text{plant gain at 50 kHz} &= A_{dc} \cdot (f_P / f_{co}) = 20 \cdot (1447 / 50{,}000) \\
&= 0.579 \text{ V/V} = -4.74 \text{ dB}
\end{aligned}
$$

with a phase contribution of approximately −90 degrees from the dominant pole.

5.3 Choosing the Crossover Frequency

The crossover frequency (where loop gain equals unity) sets the trade-off between transient response and noise immunity. A higher crossover gives faster response to load steps but risks amplifying switching noise and approaching the limits of the small-signal model. The standard guideline is to place crossover between one-tenth and one-fifth of the switching frequency. At fsw = 500 kHz, this gives a usable range of 50 kHz to 100 kHz. The lower end, 50 kHz, is chosen to keep a comfortable margin below the switching frequency and to stay safely within the validity of the averaged model, while still being fast enough to meet the ±200 mV transient specification.

5.4 Type II Compensator Design

A Type II compensator provides high DC gain (for accurate regulation through its integrator action), a zero to restore phase near crossover, and a high-frequency pole to attenuate switching noise. It is built around the error amplifier with R_COMP, C_COMP, and C_HF on the COMP pin. The three components are designed in turn.

R_COMP — sets crossover gain. This resistor sets the mid-band gain of the compensator, and therefore the crossover frequency. The datasheet design equation is the product of two terms:

$R\_COMP = [2\pi \cdot f\_co \cdot C\_out / g\_mps] \cdot [Vout / (V\_REF \cdot g\_mea)]$

The first term is the transconductance gain the loop needs at crossover:

term A = $2\pi \cdot 50{,}000 \cdot 66\mu F / 12 = 20.73 / 12 = 1.728$

The second term is the internal scaling from the feedback divider ratio and the error-amplifier transconductance (g_mea = 350 µA/V, V_REF = 0.8 V):

term B = $5 / (0.8 \cdot 350\mu A/V) = 5 / 2.80e\text{-}4 = 17{,}857$

Multiplying:

$R\_COMP = 1.728 \cdot 17{,}857 = 30{,}858\ \Omega \rightarrow 30.9\ k\Omega$ (E96)

C_COMP — places the compensator zero. The compensator zero is positioned to cancel the plant's dominant pole at 1447 Hz. Cancelling the pole with the zero flattens the loop's phase in the crossover region, restoring phase margin. The value is:

$C\_COMP = 1 / (2\pi \cdot R\_COMP \cdot f\_P)$
$= 1 / (2\pi \cdot 30{,}900 \cdot 1447)$

Building the denominator: $2\pi \cdot 30{,}900 = 194{,}150$; then $\times$ 1447 = 2.81e8. So:

$C\_COMP = 1 / 2.81e8 = 3.56\ nF \rightarrow 3.3\ nF$ (standard value)

With the standard 3.3 nF part, the actual zero lands at:

$f\_z,c = 1 / (2\pi \cdot 30{,}900 \cdot 3.3nF) = 1560\ Hz$

This sits essentially on top of the 1447 Hz plant pole, achieving the intended cancellation.

C_HF — places the high-frequency pole. A small capacitor in parallel adds a high-frequency pole that rolls off gain past crossover to reject switching noise. The datasheet gives two candidate formulas and instructs taking the larger result:

Option A (cancels ESR zero):
$C_{HF,A} = C_{out} \cdot R_{ESR} / R_{COMP} = (66\mu F \cdot 1.5m\Omega) / 30{,}900 = 3.2\ pF$

Option B (pole at fsw/2):
$C_{HF,B} = 1 / (\pi \cdot R_{COMP} \cdot f_{sw}) = 1 / (\pi \cdot 30{,}900 \cdot 500{,}000) = 20.6\ pF$

Taking the larger and rounding up to the next standard value gives C_HF = 22 pF (C0G dielectric for stability). The resulting pole is:

$f_{p,c} = 1 / (2\pi \cdot 30{,}900 \cdot 22pF) = 234\ kHz$

At 234 kHz this is 4.7× above the 50 kHz crossover, so it shapes the response well after unity gain and does not erode phase margin.

5.5 Loop Verification

Two independent checks confirm the design meets its targets.

Hand crossover check. Between its zero and pole, the compensator's gain is approximately $R_{COMP} \cdot g_{mea}$, scaled by the feedback divider attenuation ($V_{REF}/V_{out} = 0.16$):

$$
\begin{aligned}
|G_c|_{mid} &= R_{COMP} \cdot g_{mea} \cdot (V_{REF}/V_{out}) \\
&= 30{,}900 \cdot 350\mu A/V \cdot 0.16 \\
&= 10.8 \cdot 0.16 \\
&= 1.73\ V/V
\end{aligned}
$$

Multiplying by the plant gain at 50 kHz from 5.2:

$|T|$ at $f_{co} = 1.73 \cdot 0.579 = 1.00\ V/V = 0\ dB$ ✓

The loop gain is exactly unity at 50 kHz, confirming the design crosses over where intended.

Bode plot verification. The full loop transfer function (plant × compensator) was evaluated numerically in Python across frequency. The resulting Bode plots (Appendix B, scripts plant_bode.py and loop_bode.py) confirm:

- Crossover frequency ≈ 47–50 kHz

- Phase margin ≈ 62 degrees
- Gain margin large (a Type II loop does not reach −180° within the band)

A phase margin of 62 degrees is comfortably within the 45–70 degree range considered well-damped, predicting a transient response with minimal ringing and no risk of instability. This satisfies the stability requirement without physical hardware, which is the validation approach taken for this paper design (see Section 10).

5.6 Compensation Component Summary

Ref	Component	Value	Part Number	Sets
R_COMP	Comp resistor	30.9 kΩ 1%	Vishay CRCW060330K9FKEA	Crossover gain
C_COMP	Comp cap	3.3 nF X7R	Murata GRM188R71H332KA01D	Zero (1.56 kHz)
C_HF	HF cap	22 pF C0G	Murata GRM1885C1H220JA01D	HF pole (234 kHz)

6. Schematic

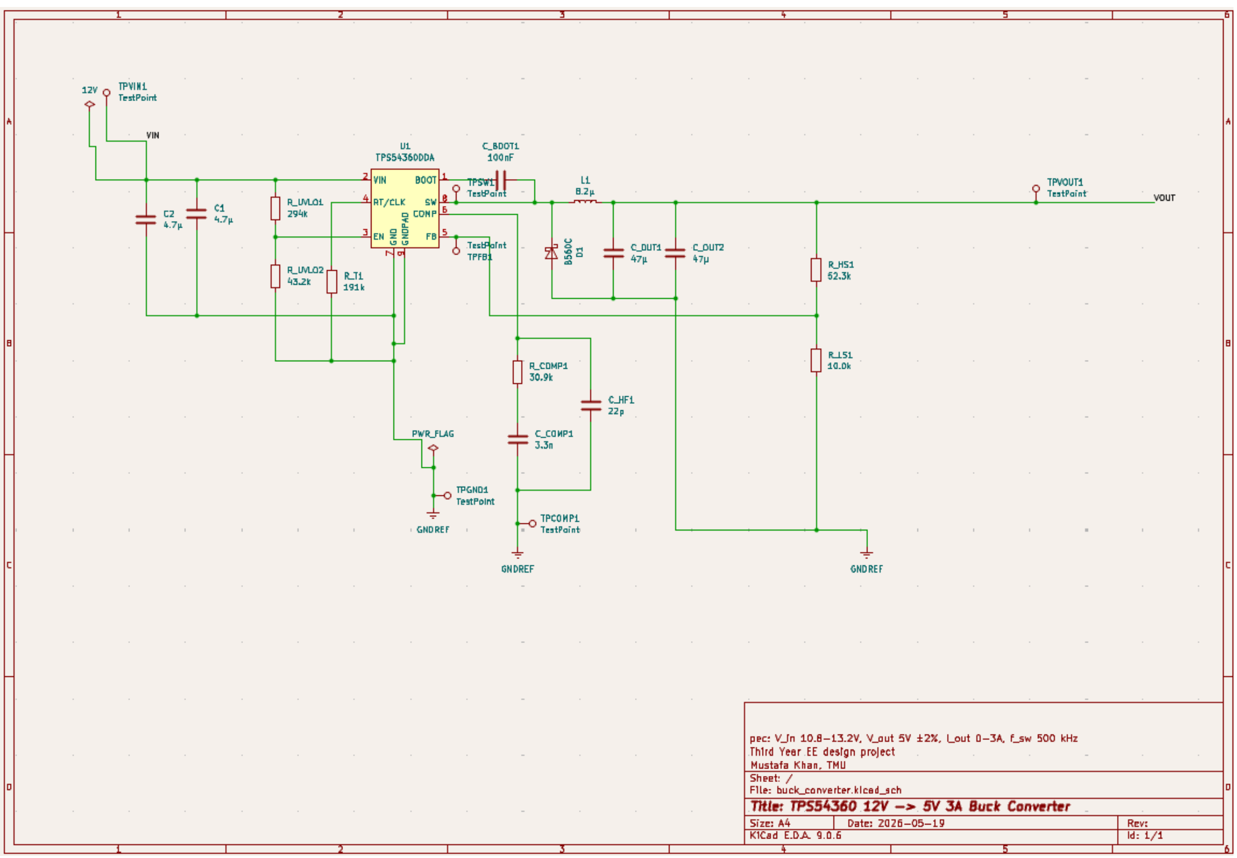


Figure 1: Complete schematic of the TPS54360-based 12V→ 5V, 3A asynchronous buck converter. The design passed Electrical Rules Check (ERC) with zero errors and zero warnings.

The complete converter schematic is shown in Figure 1. The circuit centers around the TPS54360 controller (U1). The surrounding components are grouped by function as follows.

The power stage forms the main energy path. Input capacitors C_IN1 and C_IN2 connect to the VIN pin to supply the pulsed input current drawn during each switching cycle. The integrated high-side MOSFET switches the SW pin, which connects to inductor L1 feeding the output. The Schottky catch diode D1 is placed from SW to ground, with its cathode facing the switch node. This provides the inductor current path when the high-side switch is off. Output capacitors C_OUT1 and C_OUT2 filter the inductor ripple, creating a smooth 5 V output.

The bootstrap capacitor C_BOOT connects between the BOOT and SW pins. It generates the elevated gate-drive voltage required by the high-side N-channel MOSFET. It charges while the switch node is low and then rides up with it when the node swings high.

The feedback network sets the output voltage. The divider formed by R_HS and R_LS scales the 5 V output down to the 0.8 V internal reference at the FB pin. This establishes the regulation point. These are 1% resistors since output accuracy depends directly on the divider ratio.

The switching frequency is set by R_T, which connects from the RT/CLK pin to ground, programming the 500 kHz operating frequency. It is placed close to the pin because the datasheet identifies it as noise-sensitive.

The enable and undervoltage-lockout function is managed by the divider R_UVLO1 and R_UVLO2 on the EN pin. This keeps the converter off until the input rises to 9 V. It also allows it to run down to 8 V, providing 1 V of hysteresis to prevent chattering near the threshold.

The compensation network, which includes R_COMP, C_COMP, and C_HF, connects to the COMP pin. This Type II network shapes the control loop for a 50 kHz crossover with a 62° phase margin. Its layout remains short and is routed away from the SW node to avoid coupling switching noise into the sensitive feedback path.

The schematic was verified using KiCad's Electrical Rules Check. The check reported zero errors and zero warnings, confirming a complete and electrically valid design with no floating pins or unconnected nets.

Drop your schematic image in as Figure 1 above the body text or right after the first sentence. One check before exporting the image: ensure D1's orientation in your actual schematic matches what the text states (cathode to SW). Confirm the final version is correct before the caption commits you to it.

7. PCB Layout

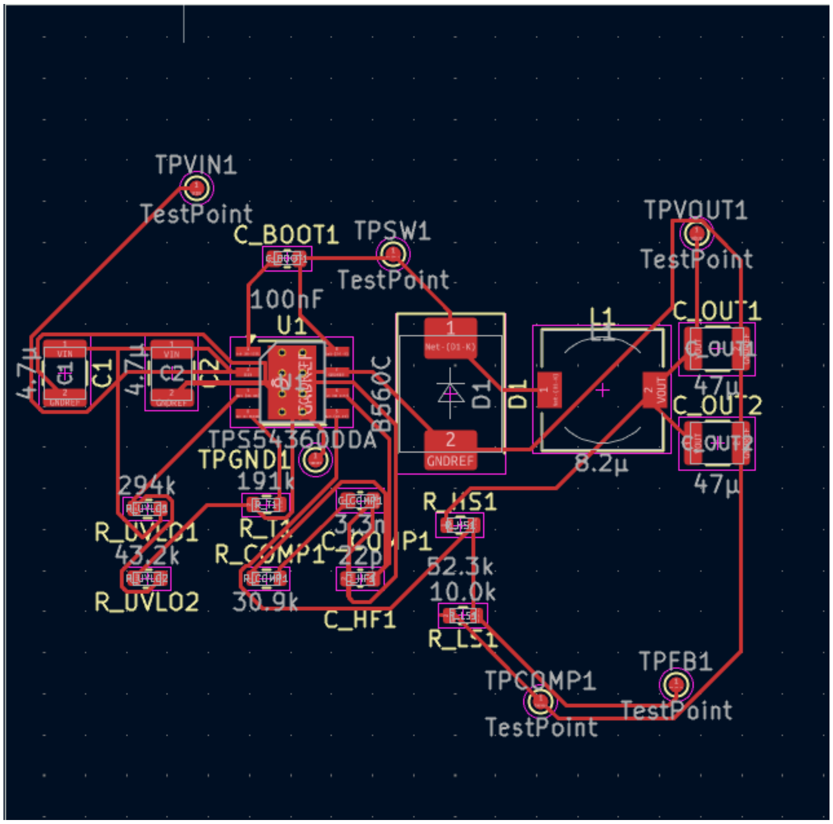


Figure 2: Top copper layer showing component placement. The input capacitors, controller (U1), and Schottky diode (D1) are grouped to minimize the high-frequency switching loop, with the inductor placed close to the switch node.

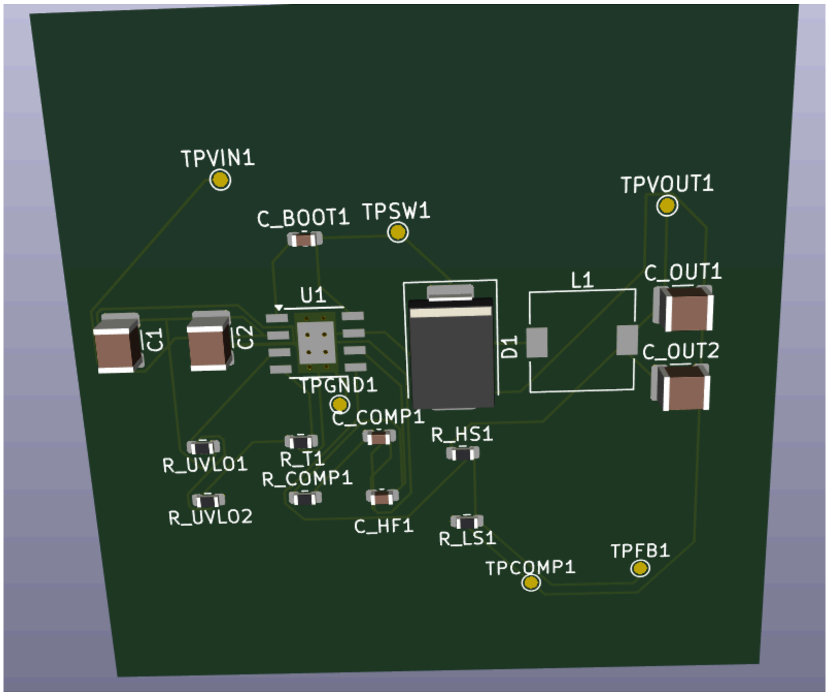


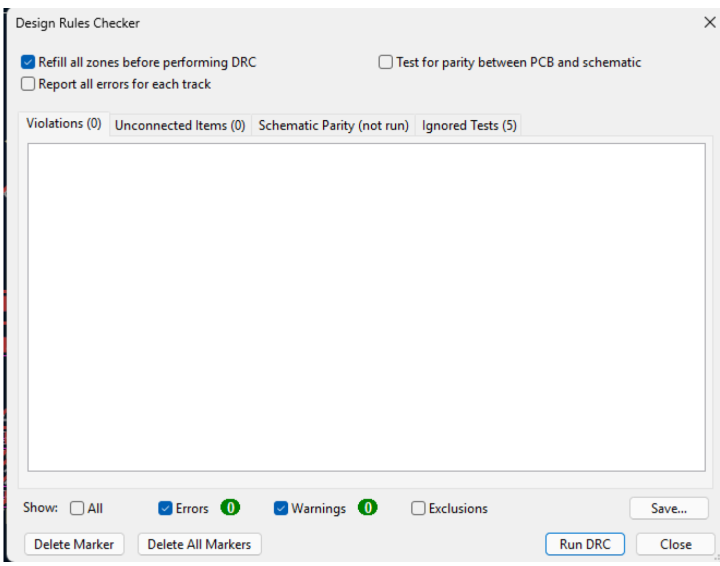
Figure 3: 3D render of the assembled board.


Figure 4: KiCad Design Rule Check result, showing zero errors.

The PCB is a four-layer board, measuring about 50 × 50 mm. It has a layer stack that includes signal/components on top, a solid ground plane on the second layer, a power layer, and signals on the bottom. Choosing four layers instead of two is intentional. A continuous internal ground plane beneath the switching components is the most effective way to control noise and provide a low-impedance return path in a switching converter.

The main goal of the layout is to minimize the high-frequency switching loop. This loop carries the pulsed current flowing through the input capacitor, the high-side MOSFET, the Schottky diode, and back to the input capacitor with each switching cycle. Because this current switches on and off quickly, any loop inductance produces voltage spikes and radiated noise. Therefore, the input capacitors, U1, and the Schottky diode D1 are placed as close together as possible to keep this loop tight. D1 is positioned right next to U1 to shorten the connections from the anode to ground and the cathode to SW.

The switch node, SW, is kept as small as practical. This is the noisiest net on the board because it swings the full input voltage at the switching frequency. It is given enough copper to carry the current without acting as an antenna. The inductor is placed close to the SW pin to maintain a compact node.

The feedback and compensation nets are routed away from the switch node. The FB and COMP pins have high impedance and are sensitive to noise. If switching noise couples into them, it can disrupt regulation and reduce loop stability. These traces are kept short and routed in a different area of the board from SW. When they must run near other signals, the ground plane underneath offers shielding.

Thermal management is handled with copper pour and vias. The TPS54360's thermal pad connects to the internal ground plane through an array of thermal vias beneath U1, spreading heat into the larger copper area. A similar array of vias is placed under the Schottky diode D1, which dissipates the most power (0.91 W at full load). The top and bottom copper layers include ground pours, which further increase thermal mass and help with heat spreading.

Grounding follows a single-plane strategy. The solid internal ground layer provides a continuous return path for all currents. This avoids ground loops and minimizes impedance between the power ground and the signal ground. The power ground, which includes high-current input and diode return, connects to the common plane with the analog ground, which includes the feedback divider and compensation. The analog components are positioned so their return currents do not flow through the noisy switching-loop area.

The completed layout passed KiCad's Design Rule Check (DRC) with zero errors. This confirms that all clearances, trace widths, and connections meet the specified design rules.

This section accounts for every significant source of power loss in the converter at full load (5 V, 3 A, 15 W output) and uses the total to predict efficiency. The losses are derived from the operating point and component parameters established in Sections 4 and 5. Where a loss depends on a datasheet curve that is not available in closed form (core loss, gate charge), the value is estimated and flagged as such, so the budget is transparent about which figures are rigorous and which are approximate.

8.1 Dominant Losses (Calculated)

Schottky diode conduction. The catch diode conducts the inductor current during the off-time, a fraction $(1 - D)$ of each cycle. With a forward voltage of 0.55 V at 3 A:

$P_D = I_{out} \cdot V_F \cdot (1 - D) = 3 \cdot 0.55 \cdot 0.55 = 0.908$ W

This is the single largest loss in the converter — over half the total — and is the direct consequence of the asynchronous topology. A synchronous low-side FET would replace this diode drop with a much smaller I^2R term.

High-side MOSFET conduction. The integrated high-side FET conducts during the on-time (fraction D), dissipating I^2R loss across its on-resistance of 92 mΩ:

$$P_{cond} = I_{out}^2 \cdot R_{DS(on)} \cdot D = 3^2 \cdot 0.092 \cdot 0.450 = 0.373 \text{ W}$$

High-side MOSFET switching. Each switching transition dissipates energy as voltage and current overlap during the finite rise and fall times. With an estimated combined transition time of roughly 20 ns:

$$\begin{aligned} P_{sw} &= 0.5 \cdot V_{in} \cdot I_{out} \cdot (t_{rise} + t_{fall}) \cdot f_{sw} \\ &= 0.5 \cdot 12 \cdot 3 \cdot 20\text{ns} \cdot 500{,}000 \\ &= 0.180 \text{ W} \end{aligned}$$

Inductor DC resistance. The inductor's winding resistance of 19.1 mΩ dissipates I^2R loss based on the RMS inductor current of 3.01 A:

$$P_{DCR} = I_{L,RMS}^2 \cdot DCR = 3.01^2 \cdot 0.0191 = 0.173 \text{ W}$$

8.2 Minor Losses (Estimated)

The remaining losses are individually small (each roughly 1–3% of the total) and depend on parameters not available in closed form, so they are estimated:

- **Inductor core loss** ≈ 0.050 W — frequency- and flux-dependent; estimated from the core material class, as the manufacturer's core-loss curve is not given in closed form.
- **Gate-drive loss** ≈ 0.018 W — the energy to charge and discharge the internal FET gate each cycle ($Q_g \cdot V_{drive} \cdot f_{sw}$).
- **Controller quiescent** ≈ 0.018 W — the chip's own operating current.
- **Capacitor ESR loss** ≈ 0.005 W — I^2R in the input and output capacitor ESR; negligible because ceramic ESR is in the low-milliohm range.

8.3 Total and Efficiency

Loss source	Power (W)	% of total	Basis
Schottky conduction	0.908	51	Calculated
HS FET conduction	0.375	21	Calculated
HS FET switching	0.180	10	Calculated
Inductor DCR	0.173	10	Calculated

Inductor core	0.050	3	Estimated
Gate drive	0.018	1	Estimated
Controller quiescent	0.018	1	Estimated
Capacitor ESR	0.005	<1	Estimated
Total	**1.73**	**100**	

The efficiency at full load follows directly:

$$\eta = P_{out} / (P_{out} + P_{loss}) = 15 / (15 + 1.73) = 89.7\%$$

This meets the design target of greater than 90% within the accuracy of the estimate, and the budget makes clear where the headroom is: the Schottky diode alone accounts for more loss than every other source combined. The efficiency versus load current is plotted in Figure 6, which shows efficiency rising as load increases (fixed losses become proportionally smaller) before the I^2R conduction terms begin to dominate at the high end.

Figure 6. Calculated efficiency versus output current at 12 V input. (Plot generated by efficiency_plot.py — see Appendix B.)

9. Bill of Materials

The complete bill of materials is listed below. Resistors are 1% metal-film 0603, and capacitors are ceramic in the dielectric and package noted. Unit prices are approximate and should be confirmed against a distributor (Digi-Key, Mouser) at the time of ordering, as component pricing varies.

Ref	Qty	Component	Value / Rating	Manufacturer P/N	Approx. unit (CAD)
U1	1	Buck controller	60 V, 3.5 A	TI TPS54360DDA	4.50
L1	1	Power inductor	8.2 µH, 8.9 A sat	Coilcraft XAL7030-822MEC	2.80
C_IN1, C_IN2	2	Input cap	4.7 µF 25V X7R 1210	Murata GRM32ER71E475KA88L	0.45

C_OUT1,C_OUT2	2	Output cap	47 µF 16V X5R 1210	Murata GRM32ER61C476ME15L	0.70
C_BOOT	1	Bootstrap cap	100 nF 25V X7R 0603	Murata GCM188R71E104KA37D	0.15
C_COMP	1	Comp cap	3.3 nF X7R 0603	Murata GRM188R71H332KA01D	0.15
C_HF	1	Comp HF cap	22 pF C0G 0603	Murata GRM1885C1H220JA01D	0.15
D1	1	Schottky diode	60 V, 5 A	Diodes Inc B560C-13-F	0.55
R_HS	1	FB top resistor	52.3 kΩ 1% 0603	Vishay CRCW060352K3FKEA	0.10
R_LS	1	FB bottom resistor	10.0 kΩ 1% 0603	Vishay CRCW060310K0FKEA	0.10
R_T	1	Freq-set resistor	191 kΩ 1% 0603	Vishay CRCW0603191KFKEA	0.10
R_UVLO1	1	UVLO top resistor	294 kΩ 1% 0603	Vishay CRCW0603294KFKEA	0.10
R_UVLO2	1	UVLO bottom resistor	43.2 kΩ 1% 0603	Vishay CRCW060343K2FKEA	0.10
R_COMP	1	Comp resistor	30.9 kΩ 1% 0603	Vishay CRCW060330K9FKEA	0.10
PCB	1	4-layer board	~50 × 50 mm	(fab house)	~10.00

Approximate component subtotal (excluding PCB): under CAD 13 in single-unit quantities. The controller and inductor dominate the cost.

10. Validation Approach

This is a complete paper design: schematic, component selection, control-loop compensation, and PCB layout, verified by analysis rather than by building and bench-testing a physical prototype. The validation therefore rests on three independent layers, each checking a different aspect of the design.

First-principles calculation. Every component value is derived from the governing datasheet equations and the established operating point, with worst-case conditions used throughout (maximum input voltage for ripple, minimum input for RMS current, full load for losses). Each result is checked for adequate margin against component ratings — for example, the inductor saturation current at roughly 2× the chip current limit, and the minimum on-time at 6.1× margin. These calculations are collected in Appendix A.

Numerical control-loop analysis. The stability of the feedback loop, which cannot be confirmed by static calculation alone, is verified by evaluating the full loop transfer function (plant × compensator) across frequency in Python. The resulting Bode plots confirm a crossover near 50 kHz with approximately 62° of phase margin, predicting a well-damped response with no risk of oscillation. An independent hand calculation confirms the loop gain equals unity at the target crossover frequency. These scripts and plots are in Appendix B.

Electrical and physical rule checks. The schematic was verified with KiCad's Electrical Rules Check (ERC), which reported zero errors and zero warnings, confirming no floating pins or missing connections. The PCB layout was verified with the Design Rule Check (DRC), confirming all clearances, trace widths, and connections meet the specified constraints.

The limitation of this approach is that it cannot capture real-world effects that only appear in hardware: actual thermal behavior under load, parasitic inductance in the switching loop, EMI, and component tolerance stack-up. These would be the subject of bench validation, discussed as future work in Section 11. The analysis-based approach is nonetheless sufficient to demonstrate that the design is sound and that all components are correctly sized with documented margin.

11. Limitations and Future Work

No physical prototype. The most significant limitation is that the design has not been built and tested. A bench prototype would allow measurement of actual efficiency across load, output ripple and transient response on an oscilloscope, and thermal imaging under sustained full load. The ENG 303 laboratory provides the oscilloscope, power supply, multimeter, and function generator needed for this validation, making a built prototype a realistic next step. The function generator in particular would allow direct measurement of the loop frequency response to confirm the predicted phase margin against hardware.

Asynchronous topology limits efficiency. The loss budget shows the Schottky diode dissipating 0.91 W, over half the total loss. Converting to a synchronous design — replacing the diode with a low-side MOSFET, using a controller such as the TPS54360's synchronous siblings — would cut this loss substantially and push efficiency from roughly 90% toward 95%. This is the single highest-impact design change available.

Fixed operating point. The converter is designed for a narrow 12 V nominal input. Widening the input range (for example, to a 9–36 V automotive range) and adding features such as adjustable soft-start, power-good indication, and output overvoltage protection would make the design more broadly applicable.

Estimated minor losses. The core loss, gate-drive loss, and quiescent terms are estimated rather than derived from measured curves. Bench measurement of total input power would resolve the true value of these terms and validate the efficiency prediction.

Future work, in priority order: build and bench-test the prototype; redesign as a synchronous converter and compare measured efficiency; extend the input range and add protection features.

12. References

1. Texas Instruments, *TPS54360 60-V Input, 3.5-A, Step-Down DC-DC Converter with Eco-mode*, datasheet SLVSBB4G. (Primary design reference; all design equations cited by number refer to this document.)
2. Coilcraft, *XAL7030 Series High-Current Shielded Power Inductors*, datasheet.
3. Murata, capacitor datasheets for GRM32ER61C476ME15L, GRM32ER71E475KA88L, GCM188R71E104KA37D, GRM188R71H332KA01D, and GRM1885C1H220JA01D.
4. Diodes Incorporated, *B560C Surface-Mount Schottky Barrier Rectifier*, datasheet.
5. R. W. Erickson and D. Maksimović, *Fundamentals of Power Electronics*, 3rd ed., Springer. (Buck topology, current-mode control, and loss modeling.)
6. Texas Instruments, application notes on peak current-mode control and Type II compensation design (SLVA series).
7. Vishay, *CRCW e3 Thick Film Chip Resistors* datasheet (feedback, frequency-set, UVLO, and compensation resistors).